



Using technology to say "approved" more often. An automated loan approval system designed to serve the unbanked and underbanked with speed and fairness.

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KweyuV	Add link to live heroku application	53a4098 · 1 minute ago	🕒
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📖 README ✎ ☰



Digitizing Loan Approvals for Financial Inclusion: A Machine Learning Approach for Kenyan Microfinance Institutions (MFI's) and Digital Lenders

Business Problem

Kenyan microfinance institutions (MFIs) and digital lenders face challenges in automating loan approvals, reducing manual delays, and improving fairness for underserved groups. This project delivers a machine learning solution that predicts loan approval using both financial and behavioral (sentiment) data, supporting digital transformation and financial inclusion.

Solution Overview

- **Automated loan approval** using XGBoost and sentiment analysis
- **Sentiment features** extracted from applicant descriptions using VADER
- **Robust feature engineering**: loan-to-income ratio, debt-to-income ratio, outlier capping
- **Business insights**: visualizations and explainable predictions

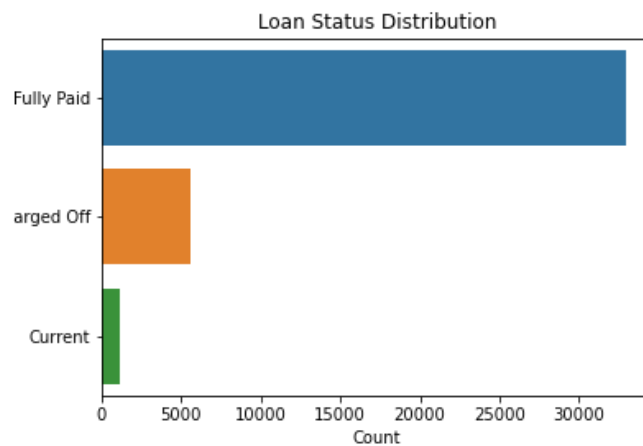
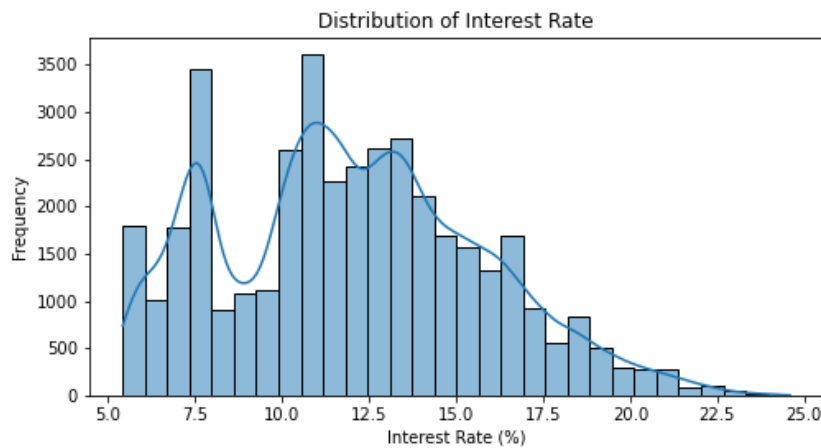
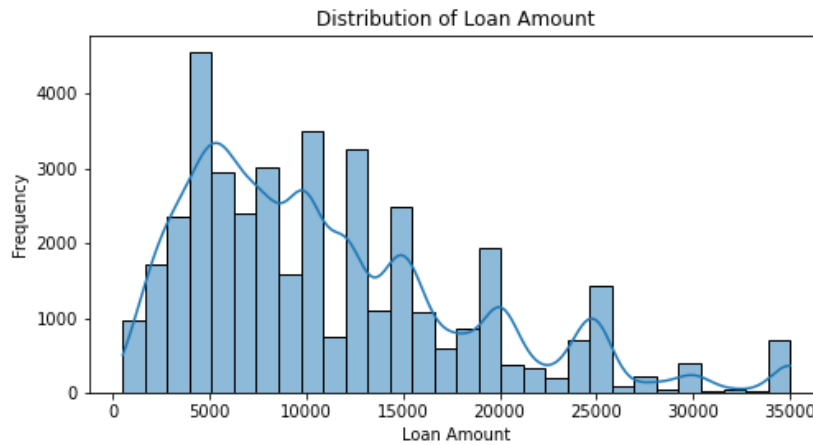
Data & Features

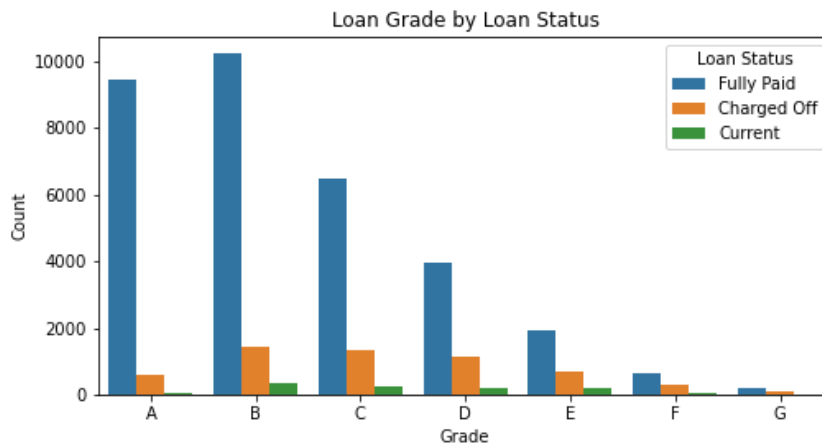
- **Source**: Real-world loan application data
- **Key features**: loan amount, term, interest rate, grade, employment length, home ownership, annual income, purpose, loan-to-income ratio, DTI, sentiment score
- **Target**: Loan approval status (Approved/Not Approved)

Workflow

1. Data Cleaning & EDA

- Drop columns with excessive missing values
- Visualize distributions (loan amount, interest rate, loan status, loan grade by loan status, loan amount by loan status)
- Example:



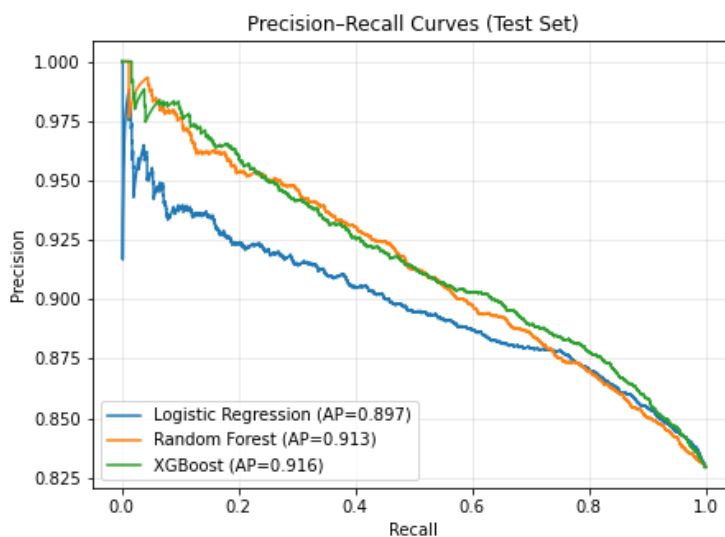


2. Feature Engineering

- Create ratio features (loan-to-income, DTI)
- Encode categorical variables
- Cap outliers
- Integrate sentiment features:
 - VADER compound score from applicant description
 - Sentiment mapped to positive/neutral/negative

3. Modeling

- Train/test split
- Handle class imbalance with SMOTE
- Models - Logistic Regression, Random Forest, XGBoost, Stacking ensemble
- Evaluate with ROC-AUC, PR-AUC, F1-score



4. Deployment

- Flask API for real-time predictions
- Heroku deployment guide
- `/predict` endpoint accepts JSON and returns approval decision, probability, and sentiment analysis

Sentiment Analysis Impact

- Sentiment features help capture applicant optimism, distress, or intent
- Rule-based override: highly negative sentiment can auto-reject applications
- Business value: improves fairness, flags risky applications, and supports responsible lending

How to Run the Entire Notebook

To execute all cells in the notebook and generate all outputs and visualizations:

1. Open `Loan_approval_final.ipynb` in Jupyter Notebook, JupyterLab, or VS Code.
2. Select "Run All" from the menu (usually under `Cell > Run All` or the run-all icon).
3. Wait for all cells to finish executing. This will:
 - Process the data
 - Engineer features
 - Train and evaluate models
 - Generate and save all plots to the `images/` folder
 - Output predictions and explanations

Tip: If you encounter any errors, make sure all dependencies are installed (see the section on installing dependencies) and restart the kernel before running all cells again.

Here is a link to the live application :

Releases

No releases published

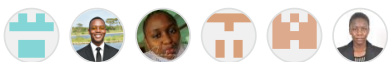
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Packages

No packages published

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Languages

● Jupyter Notebook 100.0%